

ON THE COMPLEXITY OF THE
EXTENDED STRING-TO-STRING
CORRECTION PROBLEM*

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ABSTRACT

The Extended String-to-String Correction Problem [ESSCP] is defined as the problem of determining, for given strings A and B over alphabet V, a minimum-cost sequence S of edit operations such that $S(A) = B$. The sequence S may make use of the operations: Change, Insert, Delete and Swap, each of constant cost W_C , W_I , W_D , and W_S respectively. Swap permits any pair of adjacent characters to be interchanged.

The principal results of this paper are:

- (1) a brief presentation of an algorithm (the CELLAR algorithm) which solves ESSCP in time $O(|A|*|B|*|V|^s * s)$, where $s = \min(4W_C, W_I + W_D)/W_S + 1$;
- (2) presentation of polynomial time algorithms for the cases
 - (a) $W_S = 0$, (b) $W_S > 0$, $W_C = W_I = W_D = \infty$;
- (3) proof that ESSCP, with $W_I < W_C = W_D = \infty$, $0 < W_S < \infty$, suitably encoded, is NP-complete.

(The remaining case, $W_S = \infty$, reduces ESSCP to the string-to-string correction problem of [1], where an $O(|A|*|B|)$ algorithm is given.) Thus, "almost all" ESSCP's can be solved in deterministic polynomial time, but the general problem is NP-complete.

1. Introduction

The Extended String-to-String Correction Problem [ESSCP] can be defined as follows:

Given: Two strings A and B over a finite alphabet V, and non-negative integers k, W_C , W_I , W_D , and W_S

Property: There exists a sequence S of single character edit operations chosen from the set $\{(a \rightarrow b) \mid a, b \in V \cup \{\lambda\}\} \cup \{(ab \rightarrow ba) \mid a, b \in V\}$ such that $A \xrightarrow{S} B$, and $\sum_{s \in S} W(s) \leq k$.

* Work reported herein was supported by NSF Grant Number GJ-33014.

If $a, b \in V$, $W((a \rightarrow b)) = \begin{cases} W_C & \text{if } a \neq b \\ 0 & \text{otherwise,} \end{cases}$

$W((\lambda \rightarrow a)) = W_I$, $W((a \rightarrow \lambda)) = W_D$,

and $W((ab \rightarrow ba)) = W_S$.

The relation $X \xrightarrow{(u \rightarrow v)} Y$ holds, iff $\exists x, y \in V^*$ such that $X = xuy$, and $Y = xvy$. If

$S = s_1 s_2 \dots s_n$ is a sequence of edit operations, $X \xrightarrow{S} Y$ iff $X_0, \dots, X_n \in V^*$ exist such that $X = X_0$, $X_{i-1} \xrightarrow{s_i} X_i$ for $i = 1, \dots, n$, and $X_n = Y$.

The operation $(\lambda \rightarrow a)$ permits insertion of a single character into a string, and has weight W_I ; $(a \rightarrow \lambda)$ permits deletion of a single character; $(a \rightarrow b)$ represents changing one character to another; and $(ab \rightarrow ba)$ permits swapping or interchanging the positions of two adjacent characters of a string. We will frequently refer to these operations as Insertion, Deleteion, Change, and Swap with weights W_I , W_D , W_C , and W_S , respectively.

Wagner and Fischer [1] have studied a restricted version of this problem, in which $W_S = \infty$ (or, equivalently, Swap operations are not permitted). There, a graph model of the problem, called a trace, was introduced, and the problem was shown to be solvable on a random access computer in time $O(|A|*|B|)$. Lowrance and Wagner [2] studied the "extended" problem in which Swap operations are permitted. They extended the notion of trace slightly, and showed that, for given strings A and B, corresponding to any sequence S of edit operations such that $A \xrightarrow{S} B$, a trace T exists such that $C(T) \leq W(S)$, and that, for each trace T from A to B a sequence S of edit operations exists such that $C(T) = W(S)$. Lowrance and Wagner [2] also presents

proof that whenever $2W_S \geq W_I + W_D$, the extended string-to-string correction problem still can be solved in time $O(|A| * |B|)$. This result generalizes, via an algorithm of Wagner and Brown [3] called the "CELLAR algorithm" to show that, if $(i+1)W_S \geq W_I + W_D$ for some integer i , then the ESSCP can be solved in time $O(i * |V|^{i+1} * |A| * |B|)$.

In the present paper, we wish to show that the ESSCP is NP-complete. Specifically, we show that with weights fixed at

$$W_C = W_I = \infty, W_S = 1 \text{ and } W_D < \infty$$

(or symmetrically,

$$W_C = W_D = \infty, W_S = 1 \text{ and } W_I < \infty),$$

a deterministic polynomial-time solution of this restricted ESSCP yields a deterministic polynomial-time solution of MINIMAL SET COVER [4]. Any restrictions on the assignment of values to the W 's yields a restricted ESSCP. For such a restricted ESSCP, the input becomes A , B , and k only, and time can be measured as a function of the length of these quantities. Then the following classes of restricted ESSCP's will be shown to have deterministic polynomial-time solutions:

$$W_C < \infty, \text{ or } W_C = \infty \text{ and:}$$

$$W_S = 0, \text{ or } W_S = \infty, \text{ or } W_I = W_D = \infty, \text{ or}$$

$$W_S, W_I, W_D < \infty.$$

Thus, "most" of the restricted ESSCP's have deterministic polynomial-time solutions.

2. Polynomial-time Solutions for Restricted ESSCP's

We rely primarily on the results of [1,2,3] here. In particular, Wagner and Brown [3] present an algorithm called the "CELLAR" algorithm which solves most of the restricted ESSCP's we describe in deterministic polynomial time.

Define: An extended trace from A to B is a triple

$$[U, A, B] \text{ such that } U \subset I_{|A|} \times I_{|B|}, \text{ where } I_n = 1, 2, \dots, n, \text{ and such that}$$

$$(\mu_1, \mu_2), (v_1, v_2) \in U \text{ imply } \mu_1 = v_1 \text{ iff } v_1 = v_2.$$

Any pair $(u, v) \in U$ is termed a line of U .

If $T = [U, A, B]$ is an extended trace from A to B , its "cost", $W(T)$, is defined as the sum of:

$$\begin{aligned} & |\{(i, j) \in U \mid A < i > \neq B < j >\}| * W_C, \text{ plus} \\ & |\{i \in I_{|A|} \mid \exists (i, j) \in U\}| * W_D, \text{ plus} \\ & |\{j \in I_{|B|} \mid \exists (i, j) \in U\}| * W_I, \text{ plus} \\ & |\{((i_1, j_1), (i_2, j_2)) \mid (i_k, j_k) \in U \wedge i_1 < i_2 \wedge j_1 > j_2\}| * W_S. \end{aligned}$$

These cost components represent, respectively:

- (1) the number of characters of A changed by T , times W_C ;

- (2) the number of characters of A not touched by a line of U and hence deleted by T , times W_D ;
- (3) the number of characters of B not touched by a line of U and hence inserted by T , times W_I ; and
- (4) the number of pairs of lines of U which cross, times W_S .

Lowrance and Wagner [2] have shown that for each trace T from A to B , there exists a sequence S_T of edit operations, such that $C(S_T) = W(T)$. Furthermore, to each sequence S of edit operations such that $S(A) = B$, there corresponds a trace T_S from A to B such that $W(T_S) \leq C(S)$. Thus, the search for minimal cost sequences of edit operations can be reduced to a search for a minimal cost extended trace from A to B .

Let $T = [U, A, B]$ be a trace. For any pair of integers (i, j) , define

$$U \leq (i, j) = \{(u, v) \in U \mid u \leq i \wedge v \leq j\},$$

$$U \rightarrow (i, j) = \{(u, v) \in U \mid u \leq i \wedge v > j\},$$

$$U \leftarrow (i, j) = \{(u, v) \in U \mid u > i \wedge v \leq j\}.$$

Let $A_{U(i, j)} = \{u \mid (u, v) \in U \rightarrow (i, j)\}$, and

$$B_{U(i, j)} = \{v \mid (u, v) \in U \leftarrow (i, j)\}.$$

Then $A_{U(i, j)}$ represents the set of character positions of A touched by a line of U which crosses (i, j) from "left to right" in going from A to B , and $u \in A_{U(i, j)}$ implies $1 \leq u \leq i$. Similarly, $B_{U(i, j)}$ is the set of positions of B touched by lines which cross (i, j) such that $v \in B_{U(i, j)}$ implies $1 \leq v \leq j$. We define

$T[i, j] = [U \leq (i, j), A < 1 : i >, B < 1 : j >, A_{U(i, j)}, B_{U(i, j)}]$ to be the (i, j) cut of T . The cost, $W(T[i, j])$ will be defined to correspond with $W(T)$ when $i \geq |A|$ and $j \geq |B|$. Specifically, let $X(Q)$ be the set of unordered pairs of lines of Q which cross, and let

$$\begin{aligned} R &= U \leq (i, j) \cup A_{U(i, j)} \times \{\infty\} \cup \{\infty\} \times B_{U(i, j)}. \\ \text{Take } P_1(Q) &= \{u \mid (u, v) \in Q\} \text{ and } P_2(Q) = \{v \mid (u, v) \in Q\}. \\ \text{Then } W(T[i, j]) &= |X(R)| * W_S + (i - |R| + |B_{U(i, j)}|) * W_D + \\ &\quad (j - |R| + |A_{U(i, j)}|) * W_I + |\{(u, v) \in U \leq (i, j) \mid A < u > \\ &\quad \neq B < v >\}| * W_C. \end{aligned}$$

It can easily be shown that $W(T[0, 0]) = 0$, and that $W(T[|A|, |B|]) = W(T)$.

Furthermore, if $T^* = [U, A, B]$ is a minimum-cost trace from A to B , then any cut $T^*[i, j]$ satisfies: $W(T^*[i, j]) \leq W([R \leq (i, j), A < 1 : i >, B < 1 : j >, A_{U(i, j)}, B_{U(i, j)}])$ for all traces $[R, A, B]$. This latter result suggests a systematic procedure for constructing T^* :

construct a sequence of minimal-cost cuts $T^*[0,0], T^*[1,0], T^*[1,1], \dots, T^*[|A|, |B|]$, by adding one to either i or j at each stage. More precisely, let $K[i,j,\bar{A},\bar{B}]$ be the cost of a minimal cost cut $T^*[i,j]$ such that $A_U(i,j) = \bar{A}$ and $B_U(i,j) = \bar{B}$.

We calculate, for given (i,j) , $K[i,j,\bar{A},\bar{B}]$ for all legitimate $\bar{A}, \bar{B}$ (i.e., for all subsets of $\bar{A}$ of $I_i = \{1, \dots, i\}$, and for all subsets $\bar{B}$ of $I_j = \{1, \dots, j\}$):

Suppose $T^*[i-1,j]$ was the previous cut of the sequence.

Then $K_A = K[i,j,\bar{A},\bar{B}] = \min(K_1, K_2, K_3)$, by the following case analysis:

case: $i \in \bar{A}$. Then $A<i>$ is touched by a line

$$\text{of } U \rightarrow (i,j), \text{ and } K_1 = K[i,j,\bar{A},\bar{B}] \\ = K[i-1,j,\bar{A}-\{i\},\bar{B}] + |\bar{B}|*W_S.$$

case: $i \notin \bar{A}$. Then:

case: $A<i>$ is touched by no line of U , and

$$K_2 = K[i,j,\bar{A},\bar{B}] = K[i-1,j,\bar{A},\bar{B}] + W_D;$$

case: $A<i>$ is touched by a line of $U \subseteq (i,j)$

$$\text{and } K_3 = K[i,j,\bar{A},\bar{B}] =$$

$$\min_{s \in I_j - \bar{B}} (K[i-1,j,A \cup U\{s\}] + |\{v \in \bar{B} | v < s\}|*W_S \\ + (\text{if } A<i> \neq B<s> \text{ then } W_C \text{ else } 0)).$$

A symmetric set of 3 cases holds, if $T^*[i,j-1]$ preceded $T^*[i,j]$ in the sequence of cuts, and $K[i,j,\bar{A},\bar{B}] = \min(K_A, K_B)$.

The algorithm sketched above can be shown to use only $A<\bar{A}>$ and $B<\bar{B}>$, the substrings of characters in A and B touched by the elements of $\bar{A}$ and $\bar{B}$ respectively. We can then proceed to compute $K[i,j,X,Y]$ for all substrings X of $A<1:i>$ and Y of $B<1:j>$.

The resulting algorithm, called the CELLAR algorithm, performs $\mathcal{O}(|X|+|Y|)$ operations for each distinct value of $K[i,j,X,Y]$ which it computes.

Therefore, CELLAR performs

$|A|*|B| \sum_{k=0}^{XL} |V|^{k*} \sum_{t=0}^{YL} (k+t) |V|^t = \mathcal{O}(|A|*|B|*|V|^{s*} s)$ operations where $A, B \in V^*$, and $s = XL + YL$ is the maximum $|\bar{A}|+|\bar{B}|$ calculated. We can show that s is bounded by a function of W_C, W_I, W_D and W_S , in any minimal-cost trace from A to B , independent of $|A|+|B|$. For if $T^* = [U, A, B]$, define

$$k_t = \max_{(i,j) \in S_t} |U \rightarrow (i,j)| + |U \leftarrow (i,j)|, \text{ where}$$

S_t is any sequence of integer pairs starting at $(0,0)$, ending at $(|A|, |B|)$, and such that (i_1, j_1)

follows (i_2, j_2) in S only if $(i_1, j_1) = (i_2+1, j_2)$, or $(i_1, j_1) = (i_2, j_2+1)$.

Now consider S_1 , chosen so that $|U \leftarrow (i,j)| = 0$, for all $(i,j) \in S_1$.

$$k_1 = \max_{(i,j) \in U} |U \rightarrow (i,j)| + 1. \quad \text{But in any}$$

trace T if x lines cross a given line (i,j) of T , the cost of T can be reduced, if $xW_S > W_I + W_D$, by deleting (i,j) from T .

$$\text{Hence } k_1 - 1 \leq \frac{W_I + W_D}{W_S} \quad (\text{See [3] for proof.})$$

Now consider S_2 , chosen such that

$$0 \leq |U \rightarrow (i,j)| - |U \leftarrow (i,j)| \leq 1.$$

Such a sequence of cuts exists for any trace T . Consider any cut $(i,j) \in S_2$, with $|U \rightarrow (i,j)| = m$, $|U \leftarrow (i,j)| = n$. The $m+n$ lines of $T^*[i,j]$ generate at least $m*n$ line crossings among themselves in T^* ; an alternative trace T' exists in which $|U' \rightarrow (i,j)| = m-n$, $|U' \leftarrow (i,j)| = 0$:

Choose any subset $Q_1 \subset U \rightarrow (i,j)$ such that

$$|Q_1| = n = |U \leftarrow (i,j)|, \text{ and let } Q = Q_1 \cup U \leftarrow (i,j).$$

Let $U' = U - Q \cup P_1(Q) \cdot P_2(Q)$, where

$$X \cdot Y = \{(x,y) \in X \times Y | \# \{u \in X | u \leq x\} = \# \{v \in Y | v \leq y\}\}.$$

(We use the notation $\#S = |S|$ here to avoid confusion.)

This replaces the $2n$ lines of Q with the $2n$ lines of $R = P_1(Q) \cdot P_2(Q)$. The lines of R touch exactly the same characters of A and of B touched by lines of Q . Furthermore, no pair of lines of R cross. In contrast, every line of Q_1 crosses a line of $U \leftarrow (i,j)$, so that Q contains n^2 pairs of crossing lines. Hence U' generates at least n^2 fewer line crossings than U , and U' may generate as many as $2n$ more change operations than does U . If $n^2 W_S > 2n W_C$, $W(T') < W(T^*)$.

Hence, if T^* is a minimal-cost trace,

$$n W_S \leq 2 W_C, \text{ where } k_2 \leq 2n+1.$$

Hence, in a minimal-cost trace T^* ,

$$k_2 \leq 4 W_C / W_S + 1, \text{ independent of } A \text{ or } B.$$

Thus, correction-time is

$$\mathcal{O}(|A|*|B|*|V|^{s*} s),$$

and correction space is

$$\mathcal{O}(|A|*|B|*|V|^s), \text{ where}$$

$$s = \min(4 W_C, W_I + W_D) / W_S + 1.$$

For $W_S > 0$, and $W_C < \infty$ or $W_I + W_D < \infty$,

this gives an

$$\mathcal{O}(|A|*|B|) \text{ correction-time.}$$

CELLAR ALGORITHM:

Procedure MINSET(A,B); A:= min(A,B);

string procedure DELETE(A,i); return(~~A~~1:i-1>·~~A~~i+1:|A|>);

comment: DELETE(A,i) deletes the i-th character of string A. The notation S·T indicates concatenation of strings S and T.;

if $4 * W_C < W_I + W_D$ then begin XL:= $2 * W_C / W_S$; YL:= XL + 1 end
else begin YL:= 0; XL:= $(W_I + W_D) / W_S + 1$ end;

INF:= $(W_I + W_D + W_S + W_C) * |A| * |B|$;

for j:= 0 to |B| do

for each string X with $|X| \leq XL$ do

for each string Y with $|Y| \leq YL$ do

K[0,j,X,Y]:= INF;

end;

K[0,0,λ,λ]:= 0;

for i:= 0 to |A| do

for j:= 0 to |B| do begin

for each string X with $|X| \leq XL$ do

for each string Y with $|Y| \leq YL$ do begin

K[i+1,j,X,Y]:= K[i,j,X,Y] + W_D ;

MINSET(K[i,j+1,X,Y], K[i,j,X,Y] + W_I);

for q:= 1 to |X| do

MINSET(K[i,j, DELETE(X,q),Y],

K[i,j,X,Y] + $W_S * (q - 1) + \gamma(X < q \rightarrow B < j + 1)$);

for q:= 1 to |Y| do

MINSET(K[i+1,j,X, DELETE(Y,q)],

K[i,j,K,Y] + $W_S * (q - 1) + \gamma(A < i + 1 \rightarrow Y < q)$);

if $|X| < XL$ then

MINSET(K[i+1,j,X·A<i+1>,Y],

K[i,j,X,Y] + $W_S * |Y|$);

if $|Y| < YL$ then

MINSET(K[i,j+1,X,Y·B<j+1>],

K[i,j,X,Y] + $W_S * |X|$);

end;

Two cases exist in which s is unbounded:

(a) $W_S = 0$, and (b) $W_C = W_I + W_D = \infty$.

The case $W_S = 0$ seems trivial, for in this case we can consider A and B to be multi-sets of characters, rather than ordered strings. Then compute $X = A \cap B$, $A' = A - X$, and $B' = B - X$. The ESSCP distance $D_{ESSCP}(A,B) = D_{WF}(A',B')$, where $D_{WF}(A',B')$ is the result obtained by the polynomial-time string-to-string correction algorithm of Wagner and Fischer[1].

The case $W_C = W_I + W_D = \infty$ is slightly more interesting. Two subcases can be distinguished: $W_I = W_D = \infty$, and $W_I < \infty$, or symmetrically, $W_D < \infty$.

We will see in the next section that

$W_I < \infty = W_D = W_C$, or symmetrically, $W_D < \infty = W_I = W_C$ leads to an NP-complete problem; but by means of a totally different algorithm, the EXXCP problem restricted to $W_S > 0$, $W_C = W_I = W_D = \infty$ can be solved in time $O(|A| + |B|)$:

For in this case, if T is an optimal trace from A to B with $W(T) < \infty$, A must be a permutation of the characters of B . The ESSCP then asks for the least costly series of Swap operations which permute A into B . But we can show that, even when multiple instances of one character occur in A or in B , the lines of T can easily be determined:

```
PERMUTE: for j = |B| to 1 do
    push j onto T[B<j>];
    for i = 1 to |A| do begin
        j := pop(T[A<i>]);
        output line (i,j) end;
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for each line of T must be balanced. Hence the only remaining opportunity for "choice" occurs when more than one instance of a character of A appears in B . However, a theorem of Wagner [5] asserts that even this "choice" is denied us:

For two lines (i_1, j_1) and (i_2, j_2) such that $A<i_1> = A<i_2>$ cannot cross in a minimal-weight trace ("reversing" them removes at least one line-crossing, and adds nothing to the weight of a trace).

Algorithm PERMUTE, above, computes U in time $O(|A| + |B|)$, where $[U, A, B]$ is an optimal trace.

3. An NP-Complete ESSCP

We will consider the system of weights

$$S = [W_S = 1, W_I = \infty, W_D < \infty, W_C = \infty],$$

and show that for each MINIMUM SET-COVER [MSC] problem g , there can be derived in polynomial-time an ESSCP, h , in this system S such that $g \in L(MSC)$ if and only if $h \in L(ESSCP_S)$.

In system S , the minimum-cost trace $T = [U, A, B]$ has the following property, if $W(T) < \infty$:

For each $j \in I_{|B|}$ there exists $(i, j) \in U$ such that $A<i> = B<j>$.

Also, if character c appears i times in B and j times in A , exactly $(j-i)$ occurrences of c must be deleted from A , if $W(T) < \infty$. Thus the deletion-cost is constant, regardless of W_D , and regardless of which positions of A are touched by lines of T .

The MSC problem can be stated:

Given finite sets $x_1, \dots, x_n$, and integer ℓ , does there exist a subset $J \subset \{1, \dots, n\}$ such that $|J| \leq \ell$, and

$$\bigcup_{j \in J} x_j = \bigcup_{j=1}^n x_j \quad ?$$

Let $w = \bigcup_{j=1}^n x_j$, let $t = |w|$ and $r = t^2$,

and choose symbols Q, R, S not in w . Take string A to be $Q^r R x_1 Q^r S^{r+1} \dots Q^r R x_n Q^r S^{r+1}$, string B to be

$$\underbrace{R Q^r \dots R Q^r}_{n \text{ times}} w \underbrace{S^{r+1} \dots S^{r+1}}_{n \text{ times}},$$

and k to be

$$(\ell+1)r - 1 + 2t(r+1)(n-1) + n(n-1)(r+1)^2 + d * W_D,$$

where $d = r * n + |x_1 \dots x_n| - |w|$.

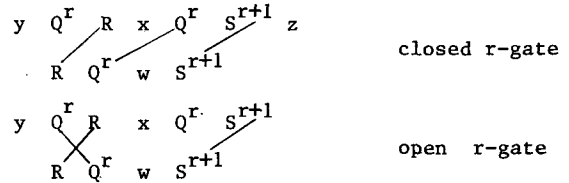
The resulting ESSCP_S is:

Given strings A and B , and integer k , does there exist a trace $T = [U, A, B]$ such that $W(T) \leq k$, for weights $W_S = 1$,

$$W_I = W_C = \infty, W_D < \infty?$$

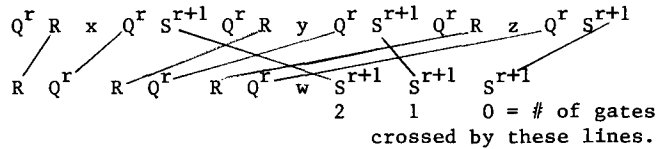
The derived ESSCP_S is satisfiable if and only if the source MSC problem is.

Our construction hinges on the object diagrammed below, which we call an r -gate:



In system S , the cost, ignoring deletion and insertion costs, of an open r -gate trace is just r , while that of a closed r -gate trace is 0. If $q \geq 1$ non-crossing lines join x to w , the cost of the resulting minimal trace is r , via the "open" r -gate trace; if 0 lines join x to w , the "closed" r -gate trace becomes optimal, at cost 0. Furthermore, each line joining w to string y or z (considered "outside" the gate shown) crosses exactly $r+1$ lines of this r -gate, independent of the "open" or "closed" trace form for this gate. Our constructed ESSCP_S essentially concatenates n r -gate "tops" onto string A , while successively nesting w into the "bottoms" of those n gates.

Example:



Note that in the "closed" positions, none of the structural Q or R -lines cross, while regardless of gate position, the S -lines each cross 0, 1, 2, ..., $n-1$ entire gates (each containing $(r+1)$ lines). Any line drawn from w must enter some one gate, and must therefore cross $(n-1)$ other gates, each containing r Q -lines, 1 R -line, and $r+1$ S -lines, a

total of $2(r+1)(n-1)$ lines crossed. Since every character of w is touched by some line, there must be a total of

$W(T) - d \cdot W_D = 2t(r+1)(n-1) + n(n-1)(r+1)^2/2 + f + gr$ line crossings in the minimal-cost trace, where g = the number of "open" gates, and f = the number of pairs of w -touching lines which cross. Since there are t w -touching lines,

$$0 < f < t(t-1)/2 < t^2 = r.$$

Thus $gr + c \leq W(T) \leq gr + c + t(t-1)/2 < (g+1)r + c$, where $c = 2t(r+1)(n-1) + n(n-1)(r+1)^2/2 + W_D \cdot d$. Hence, if and only if $W(T) \leq (\ell+1)r + c - 1 = k$, there exists a trace with ℓ or fewer open gates such that every position of w is touched by a line of T leading from w to the string inside an open gate. We obviously can identify the strings inside open gates with the sets indexed by J in the MSC problem.

4. A Generalization

The results of the previous sections can be generalized to the case where the cost of each edit operation $a \rightarrow b$, $a, b \in V \cup \{\lambda\}$ depends on a and b , $W((a \rightarrow b)) \geq 0$, while the cost of a Swap operation remains constant at W_S , no matter what characters are being transposed. (The restriction to constant Swap-cost seems necessary, if the "trace" model we rely on is to maintain its validity.) Specifically, we note that the Wagner-Fischer algorithm, the CELLAR algorithm, and PERMUTE all generalize readily to this case. Thus, if $W_C = \max_{a,b \in V} W(a \rightarrow b)$, $W_I = \max_{a \in V} W(\lambda \rightarrow a)$

and $W_D = \max_{a \in V} W(a \rightarrow \lambda)$, then:

case: $W_S = \infty$: Use Wagner-Fischer algorithm [1].

case: $0 < W_S < \infty$:

case: $W_I = W_D = W_C = \infty$: Use PERMUTE.

case: $W_I = W_C = \infty$, $W_D < \infty$, or $W_D = W_C = \infty$, $W_I < \infty$: Polynomial complete.

case: $W_C < \infty$, or $W_I + W_D < \infty$: Use the CELLAR algorithm.

case: $W_S = 0$:

The case $W_S = 0$, in the generalized-weight ESSCP, turns out to be the classical

"transportation problem" of linear programming [6]. Specifically, let $q(x, S)$ be the number of occurrences of character x in string X . Interpret w_{ij} as the number of characters "i" in A which should be changed to "j" in the optimal solution. Then solving the following transportation problem produces, as the optimal objective-function value, the cost of a minimal-weight trace T from A to B :

$$W(T) = \min \sum_{i,j \in V \cup \{\lambda\}} W(i \rightarrow j) w_{ij} \quad \text{such that}$$

$$\sum_i w_{ij} = q(j, B) \quad \text{for } j \in V$$

$$\sum_i w_{i\lambda} \leq |A|,$$

$$\sum_j w_{ij} = q(i, A) \quad \text{for } i \in V$$

$$\sum_j w_{\lambda j} \leq |B|,$$

$$w_{ij} \geq 0.$$

The solution of this problem can be computed in time $O(\binom{v+2}{2v-1})$, where $v = |V \cup \{\lambda\}|$.

The problem itself can be constructed in time

$$O(|A| + |B|).$$

Hence, when $|A| + |B| \gg |V|$, the solution-time for this case is $O(|A| + |B|)$.

References

- [1] Wagner, R. A. & Fischer, M. J., "The string-to-string correction problem", Journal of the ACM, Vol. 21, no. 1 (January 1974), pp. 168-173.
- [2] Lowrance, R. & Wagner, R. A., "An extension of the string-to-string correction problem" to be published in the Journal of the ACM.
- [3] Wagner, R. A. & Brown, T. P., "Swap-extended minimum edit distance correction for regular languages", Technical Report, Systems and Information Science Program, Vanderbilt University (January 1975).
- [4] Karp, R. M., "Reducibility among combinatorial problems" in Complexity of Computer Computations, Miller, R. E. & Thatcher, J. W. (ed.), Plenum Press (1972), pp. 85-104.
- [5] Wagner, R. A., "Generalized correction of context-free languages", Technical Report, Systems and Information Science Program, Vanderbilt University (1973).
- [6] Gaver, G. P. & Thompson, G. L., Mathematical Models: Programming and Probability (1966).